# CSST - Complication Signal Surveillance Template

Review identification, purpose and sources

SIMULATED · v2.0

| | | | |
|---|---|---|---|
| **Study / protocol**<br>GLENOS-01 / GS-2026-01 | **CIP version**<br>CIP v4.2 | **Site / code**<br>Riverside Orthopedic Center / 014 | **CSST review ID / revision**<br>CSST-014-02 / r1 |
| **Review type**<br>☐ Scheduled ☑ Triggered | **Review period / data cut-off**<br>1-31 Oct 2026; cut-off 31 Oct | **Device / procedure reviewed**<br>GLENOS reverse shoulder, all study procedures | **Previous CSST review ID**<br>CSST-014-01 (30 Sep 2026) |
| **SCML snapshot ID**<br>SCML-014, snapshot 31 Oct 2026 | **Analysis file (if used)**<br>None: counted on this form | **Prepared by / role**<br>Maria Silva / Study Coordinator | **Controlled location**<br>ISF tab 6 (Safety) / CSST-014-02 |

## A Purpose and use

The CSST is the site's review of its complications taken together. A single event is assessed in the SART and indexed in the SCML; the CSST looks at all of them at once, on the site's own schedule and after a trigger, to see whether events that each seemed explicable share a cause: the same component, the same surgeon, the same moment of follow-up or the same kind of case. A concentration of events is a question to test, not a conclusion about a device or a surgeon.

| Unit of analysis | Decision supported | Inputs | Outputs |
|---|---|---|---|
| One review of the site's complications for one period, or one review opened by a trigger. | Whether the data show no material pattern, a pattern to watch more closely, a signal to send on, or a result limited by the data. | SCML events, ICDL procedures and context, PFVR follow-up reached, SLCT surgeon sequence, SART event status and the previous CSST. | Counts by time point (C)<br>Triggers met (D)<br>Signals assessed (F)<br>Routes to the receiving records (H) |

## B Sources and their limitations

With limitation means the review goes ahead: name the records that are missing or incomplete. Not available means the counts that need that source cannot be made in this review.

| Source | Used for | ID · extraction date | Status | Limitation (records missing or incomplete) |
|---|---|---|---|---|
| **SCML snapshot** | Events and their categories | SCML-014 snapshot · 31 Oct | ☑ Complete<br>☐ With limitation<br>☐ Not available | |
| **ICDL procedure records** | Procedures, components, configuration, context | ICDL-014-001 to -008 · 31 Oct | ☐ Complete<br>☑ With limitation<br>☐ Not available | ICDL-014-005: liner lot missing; read from the explant label |
| **PFVR / visit schedule** | Participants who reached each follow-up time point | PFVR export · 31 Oct | ☐ Complete<br>☑ With limitation<br>☐ Not available | 014-0003: 6-week visit on day 52 (outside window) |
| **SLCT** | Surgeon codes and case sequence | SLCT-014 · 31 Oct | ☑ Complete<br>☐ With limitation<br>☐ Not available | |
| **SART** | Event type, seriousness and status | SART export · 31 Oct | ☑ Complete<br>☐ With limitation<br>☐ Not available | |
| **Previous CSST review** | Signals to follow up in G | CSST-014-01 · 30 Sep | ☑ Complete<br>☐ With limitation<br>☐ Not available | |

## C1 Cases and follow-up reached

These numbers are the denominators for C2 to C4. Procedures and surgeons come from the ICDL and the SLCT, enrollment from the PSEAR, follow-up from the PFVR. The time points match the columns of C2. Adapt the time points to the CIP visit schedule.

| Count | This period (n) | Previous period (n) | Change (%) | Cumulative (N) |
|---|---|---|---|---|
| Participants enrolled | 4 | 4 | 0% | 8 |
| Investigational procedures performed | 4 | 4 | 0% | 8 |
| Surgeons operating | 3 | 2 | +50% | 3 |
| Surgeons with a first study case in this period | 1 | 2 | -50% | 3 |
| Procedures within a surgeon's first 5 study cases | 3 | 4 | -25% | 7 |
| Revision procedures (component exchanged or removed) | 1 | 0 | - | 1 |
| Explanted components returned | 0 (1 pending) | 0 | - | 0 |
| Follow-up visits due | 9 | 4 | +125% | 13 |
| Follow-up visits done in window | 7 | 4 | +75% | 11 |
| Follow-up visits missed | 1 | 0 | - | 1 |
| **Participants who reached each time point (N for the columns of C2)** | | | | |
| Operated (intraoperative) | 4 | 4 | 0% | 8 |
| Reached discharge | 4 | 4 | 0% | 8 |
| Reached first postoperative visit | 4 | 4 | 0% | 8 |
| Reached 6 weeks | 3 | 2 | +50% | 5 |
| Reached 3 months | 2 | 0 | - | 2 |
| Reached 6 months | 0 | 0 | - | 0 |
| Reached 12 months | 0 | 0 | - | 0 |
| Reached 24 months | 0 | 0 | - | 0 |
| Participants withdrawn or lost to follow-up | 0 | 0 | - | 0 |

## C2 Complications by follow-up time point

Count each event once, in the time point where it occurred, against the N of that column. Reading across a row shows whether a problem appears early or grows with time.

| Event (SCML category) | Intraoperative (n / N) % | Before discharge (n / N) % | First postop visit (n / N) % | 6 weeks (n / N) % | 3 months (n / N) % | 6 months (n / N) % | 12 months (n / N) % | 24 months (n / N) % | Total (n) | Grade IIIa or higher (n) | New this period |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **N for this column (from C1)** | 8 | 8 | 8 | 5 | 2 | 0 | 0 | 0 | | | |
| **Instability / dislocation** | | | | | | | | | | | |
| Dislocation | | | | (1 / 5) 20% | | | | | 1 | 1 | 1 |
| Subluxation or instability symptoms | | | | | (1 / 2) 50% | | | | 1 | 0 | 1 |
| **Fracture** | | | | | | | | | | | |
| Intraoperative fracture | | | | | | | | | 0 | 0 | 0 |
| Periprosthetic fracture | | | | | | | | | 0 | 0 | 0 |
| Acromion or scapular spine fracture | | | | | | | | | 0 | 0 | 0 |
| **Infection / wound** | | | | | | | | | | | |
| Superficial wound problem | | (1 / 8) 12.5% | | | | | | | 1 | 0 | 0 |
| Deep or periprosthetic infection | | | | | | | | | 0 | 0 | 0 |
| Hematoma | | (1 / 8) 12.5% | | | | | | | 1 | 0 | 1 |
| **Neurologic / vascular** | | | | | | | | | | | |
| Nerve injury (axillary nerve, brachial plexus) | | | (1 / 8) 12.5% | | | | | | 1 | 0 | 0 |
| Vascular injury | | | | | | | | | 0 | 0 | 0 |
| **Device performance** | | | | | | | | | | | |
| Component loosening | | | | | | | | | 0 | 0 | 0 |
| Component disassociation (liner, glenosphere) | | | | (1 / 5) 20% | | | | | 1 | 1 | 1 |
| Implant breakage or wear | | | | | | | | | 0 | 0 | 0 |
| Radiographic change (notching, lucency, migration) | | | | | | | | | 0 | 0 | 0 |

## C2 Complications by follow-up time point (continued)

| Event (SCML category) | Intraoperative (n / N) % | Before discharge (n / N) % | First postop visit (n / N) % | 6 weeks (n / N) % | 3 months (n / N) % | 6 months (n / N) % | 12 months (n / N) % | 24 months (n / N) % | Total (n) | Grade IIIa or higher (n) | New this period |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **N for this column (from C1)** | 8 | 8 | 8 | 5 | 2 | 0 | 0 | 0 | | | |
| **Function and other** | | | | | | | | | | | |
| Stiffness or loss of motion | | | | | | | | | 0 | 0 | 0 |
| Persistent pain | | | | | | | | | 0 | 0 | 0 |
| Thromboembolic event | | | | | | | | | 0 | 0 | 0 |
| Other (SCML code): _____ | | | | | | | | | 0 | 0 | 0 |

**Overview**

| Measure | This period (n) | Previous period (n) | Change (%) | Cumulative (n / N) % | Cumulative at previous review (n / N) % |
|---|---|---|---|---|---|
| Participants with at least one complication | 3 | 2 | +50% | (5 / 8) 62.5% | (2 / 4) 50% |
| Participants with a complication of grade IIIa or higher | 1 | 0 | - | (1 / 8) 12.5% | (0 / 4) 0% |
| Participants with more than one complication | 1 | 0 | - | (1 / 8) 12.5% | (0 / 4) 0% |
| Reoperations | 1 | 0 | - | (1 / 8) 12.5% | (0 / 4) 0% |
| Revisions (component exchanged or removed) | 1 | 0 | - | (1 / 8) 12.5% | (0 / 4) 0% |
| Serious events (SAE / SADE) | 1 | 0 | - | (1 / 8) 12.5% | (0 / 4) 0% |
| Device deficiencies (all; counted apart from complications) | 1 | 0 | - | (1 / 8) 12.5% | (0 / 4) 0% |

## C3 Complications by Clavien-Dindo grade and time point

Grades come from the SCML, which copies them from the SART. Count each complication once, in the column of its time point. Device deficiencies without an adverse event are not graded.

| Grade | Intraoperative (n / N) % | Before discharge (n / N) % | First postop visit (n / N) % | 6 weeks (n / N) % | 3 months (n / N) % | 6 months (n / N) % | 12 months (n / N) % | 24 months (n / N) % | Total (n) | New this period |
|---|---|---|---|---|---|---|---|---|---|---|
| **N for this column (from C1)** | 8 | 8 | 8 | 5 | 2 | 0 | 0 | 0 | | |
| **I** · No treatment beyond simple care | | (1 / 8) 12.5% | (1 / 8) 12.5% | | (1 / 2) 50% | | | | 3 | 2 |
| **II** · Drug treatment, transfusion | | (1 / 8) 12.5% | | | | | | | 1 | 0 |
| **IIIa** · Intervention without general anesthesia | | | | | | | | | 0 | 0 |
| **IIIb** · Intervention under general anesthesia | | | | (2 / 5) 40% | | | | | 2 | 2 |
| **IVa** · Life-threatening, single organ | | | | | | | | | 0 | 0 |
| **IVb** · Life-threatening, multiorgan | | | | | | | | | 0 | 0 |
| **V** · Death | | | | | | | | | 0 | 0 |
| **Disability at discharge (suffix d)** | | | | | | | | | 0 | 0 |

## C4 Procedure context

For each marker, compare the complication rate in procedures with it and without it. Context never changes the counts in C2; it helps explain them.

| Context marker | Procedures with it (n / N) % | With it: had a complication (n / n) % | Without it: had a complication (n / n) % | Source |
|---|---|---|---|---|
| Unplanned configuration change | (1 / 8) 12.5% | (1 / 1) 100% | (4 / 7) 57% | ICDL-014-005 |
| Component substitution | (0 / 8) 0% | - | (5 / 8) 62.5% | ICDL |
| Technical deviation from the surgical technique | (0 / 8) 0% | - | (5 / 8) 62.5% | ICDL |
| Intraoperative difficulty (bone quality, deformity, revision case) | (2 / 8) 25% | (1 / 2) 50% | (4 / 6) 67% | ICDL-014-002, -007 |
| Senior support or proctoring used | (2 / 8) 25% | (1 / 2) 50% | (4 / 6) 67% | ICDL / SLCT |
| Surgeon within first ___ study cases | (7 / 8) 87.5% | (5 / 7) 71% | (0 / 1) 0% | SLCT |
| Operation outside normal hours | (1 / 8) 12.5% | (1 / 1) 100% | (4 / 7) 57% | ICDL-014-005 |
| Instrument or loaner set problem | (1 / 8) 12.5% | (1 / 1) 100% | (4 / 7) 57% | ICDL-014-002 |
| Gap in the operative record | (1 / 8) 12.5% | (1 / 1) 100% | (4 / 7) 57% | ICDL-014-005 |
| Other: _____ | | | | |

**Links between context and complications**

| Context marker | Procedures (ICDL IDs) | Complication found (C2 event · SCML ID) | Comment |
|---|---|---|---|
| Unplanned configuration change | ICDL-014-005 | Dislocation and liner disassociation (SCML-014-07, -08) | Planned +6 liner not in the set; +3 used. Same case had the weekend list and the missing lot. |
| Instrument or loaner set problem | ICDL-014-002 | Subluxation symptoms at 3 months (SCML-014-09) | Trial liner missing from the loaner set; liner chosen without a trial reduction. |
| | | | |
| | | | |

# E Comparison by surgeon, component and case type

Grades follow Clavien-Dindo (SCML). A group stands out when its rate of grade IIIa–V complications is clearly above the rate of the rest of the site. Use the surgeon codes of the SLCT and the component, lot and configuration recorded in the ICDL.

**E1 · By surgeon (SLCT code)**

| Surgeon | Procedures (n) % of all | Grade I–II (n / n) % | Grade IIIa–V (n / n) % | IIIa–V in first ___ cases (n / n) % | Note |
|---|---|---|---|---|---|
| S-01 | (3) 37.5% | (1 / 3) 33% | (0 / 3) 0% | (0 / 3) 0% | PI; experienced |
| S-02 | (3) 37.5% | (2 / 3) 67% | (0 / 3) 0% | (0 / 3) 0% | Sub-I; grade I and II only |
| S-03 | (2) 25% | (1 / 2) 50% | (1 / 2) 50% | (1 / 2) 50% | New in Oct; the only IIIb |
| | | | | | |
| | | | | | |

**E2 · By component, lot or configuration (ICDL)**

| Component · lot · configuration | Procedures (n) % of all | Grade I–II (n / n) % | Grade IIIa–V (n / n) % | Note |
|---|---|---|---|---|
| Liner 36 +3 · lot GS-L36-2607 | (2) 25% | (1 / 2) 50% | (1 / 2) 50% | Includes ICDL-014-005 |
| Liner 36 +6 | (4) 50% | (2 / 4) 50% | (0 / 4) 0% | |
| Glenosphere 36 · all cases | (8) 100% | (4 / 8) 50% | (1 / 8) 12.5% | No comparison possible |
| | | | | |
| | | | | |

**E3 · By case type**

| Case type (indication, primary or revision) | Procedures (n) % of all | Grade I–II (n / n) % | Grade IIIa–V (n / n) % | Note |
|---|---|---|---|---|
| Cuff tear arthropathy, primary | (5) 62.5% | (2 / 5) 40% | (0 / 5) 0% | |
| Fracture sequelae, primary | (2) 25% | (1 / 2) 50% | (1 / 2) 50% | Both instability events |
| Massive cuff tear with arthritis | (1) 12.5% | (1 / 1) 100% | (0 / 1) 0% | |
| | | | | |
| | | | | |

## D Signal triggers

Apply each trigger and write the value found. A trigger met opens a signal in F.

| ID | Trigger | What is compared · where | Agreed limit | Source of the limit | Value found (n / N) % | Met? |
|---|---|---|---|---|---|---|
| T1 | **Rate above the agreed limit** | Any category at any time point; one limit for all grades and a lower one for grade IIIa–V (C2, C3). | Instability: all grades > 10%; IIIa–V > 5% | Site rule, 20 Aug 2026 (POT) | All: (1 / 2) 50% at 3 mo; IIIa–V: (1 / 5) 20% at 6 wk | ☑ Yes ☐ No ☐ Not assessable |
| T2 | **Rate rising with time** | The same category grows from earlier to later time points (C2 row). | Higher at a later time point, 2+ events | Site rule, 20 Aug 2026 | Instability 0% → 20% → 50% | ☑ Yes ☐ No ☐ Not assessable |
| T3 | **Rise since the previous review** | Same category and time point, compared with the previous CSST. | Double the previous review | Site rule, 20 Aug 2026 | Instability 0 → 2 events | ☑ Yes ☐ No ☐ Not assessable |
| T4 | **New category** | A category with no earlier events now has events (C2, New this period). | Any first event in a category | Site rule, 20 Aug 2026 | Instability; device performance | ☑ Yes ☐ No ☐ Not assessable |
| T5 | **Surgeon rate** | A surgeon's rate of grade IIIa–V complications is clearly above that of the other surgeons (E1). | IIIa–V rate 2x the rest of the site, 2+ events | Site rule, 20 Aug 2026 | S-03: 50% vs 0%, but 1 event | ☐ Yes ☑ No ☐ Not assessable |
| T6 | **Component or configuration rate** | A component, lot or configuration has a grade IIIa–V rate clearly above the others (E2). | Same rule as T5 | Site rule, 20 Aug 2026 | Liner +3: 50% vs 0%, but 1 event | ☐ Yes ☑ No ☐ Not assessable |
| T7 | **Early learning cases** | Grade IIIa–V rate in a surgeon's first cases above their later cases (E1, C1). | Assessable from 6 cases per surgeon | Sponsor guidance GS-RA-01 | All surgeons have 3 cases or fewer | ☐ Yes ☐ No ☑ Not assessable |
| T8 | **Context association** | Rate with a context marker above the rate without it (C4). | Rate with marker 2x without, 2+ events | Site rule, 20 Aug 2026 | Each marker has 1 event only | ☐ Yes ☑ No ☐ Not assessable |
| T9 | **Serious events grouped** | Serious events in one category within a short window (SART, C2). | 2 serious events, same category, 90 days | CIP 11.6 | 1 SADE (dislocation) | ☐ Yes ☑ No ☐ Not assessable |
| T10 | **Device deficiencies of one type** | Repeated deficiencies of the same type or component (C2, SART). | 2 deficiencies of the same type | CIP 11.6 | 1 (liner disassociation) | ☐ Yes ☑ No ☐ Not assessable |
| T11 | **Same severe complication repeated** | The same complication of grade IIIa–V in two or more participants (C3). | Same IIIa–V complication in 2 participants | Site rule, 20 Aug 2026 | Dislocation IIIb: 1 participant | ☐ Yes ☑ No ☐ Not assessable |

These triggers are examples. Before the first inclusion, review them and set each limit with the site's clinical team and, where it applies, the sponsor; write the agreed limit and its source (CIP, sponsor guidance, site procedure). A limit changed later applies only to later reviews.

## F Assessment of each signal

One row per trigger met. Describe the pattern and the explanations still to test; do not conclude the cause. Set how the next review will show whether the response worked.

| Signal ID | Trigger (D) · events (SCML IDs) | Pattern described | Explanations to test | Disposition | What will show the response worked |
|---|---|---|---|---|---|
| S-01 | T1-T4 · SCML-014-07, -08, -09 | Two instability events in 8 procedures, both after discharge and rising at later visits. One grade IIIb (dislocation with liner disassociation, revised) after an unplanned +3 liner; one grade I (subluxation) after a loaner set without a trial liner. The only IIIb is in S-03 and a fracture-sequelae case. | ☑ Device, component or lot<br>☑ Procedure or technique<br>☑ Surgeon or learning curve<br>☐ Case complexity or participant<br>☐ Data or documentation<br>☐ Local process<br>☐ Undetermined | ☐ No material pattern<br>☐ Closer surveillance<br>☑ Sent on (see H)<br>☐ Inconclusive: data limitation | No new instability in the next 8 procedures; full liner range and trial liners in every set (ICDL); CSST monthly for 3 months. |
| | | | ☐ Device, component or lot<br>☐ Procedure or technique<br>☐ Surgeon or learning curve<br>☐ Case complexity or participant<br>☐ Data or documentation<br>☐ Local process<br>☐ Undetermined | ☐ No material pattern<br>☐ Closer surveillance<br>☐ Sent on (see H)<br>☐ Inconclusive: data limitation | |
| | | | ☐ Device, component or lot<br>☐ Procedure or technique<br>☐ Surgeon or learning curve<br>☐ Case complexity or participant<br>☐ Data or documentation<br>☐ Local process<br>☐ Undetermined | ☐ No material pattern<br>☐ Closer surveillance<br>☐ Sent on (see H)<br>☐ Inconclusive: data limitation | |
| | | | ☐ Device, component or lot<br>☐ Procedure or technique<br>☐ Surgeon or learning curve<br>☐ Case complexity or participant<br>☐ Data or documentation<br>☐ Local process<br>☐ Undetermined | ☐ No material pattern<br>☐ Closer surveillance<br>☐ Sent on (see H)<br>☐ Inconclusive: data limitation | |

# CSST - Complication Signal Surveillance Template

Earlier signals and routes

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## G Signals from earlier reviews

Check each earlier signal against the criterion set for it in F.

| Signal ID · review | Criterion set then | Result in this review | Outcome | Next step |
|---|---|---|---|---|
| S-01 · CSST-014-01 | No further nerve injury in the next 4 procedures | 0 nerve injuries in 4 procedures (Oct) | ☑ Effective<br>☐ Partly effective<br>☐ Not effective<br>☐ Not assessable | Close; keep in the routine count |
| | | | ☐ Effective<br>☐ Partly effective<br>☐ Not effective<br>☐ Not assessable | |
| | | | ☐ Effective<br>☐ Partly effective<br>☐ Not effective<br>☐ Not assessable | |
| | | | ☐ Effective<br>☐ Partly effective<br>☐ Not effective<br>☐ Not assessable | |

## H Where each signal goes

The action is followed in the receiving record; this review records only that it was sent.

| Route | Question for this review | Signal ID (F) | Receiving record · date sent | Status |
|---|---|---|---|---|
| **SLCT** | Could surgeon-dependent variation explain the pattern? | S-01 | SLCT S-03 review, 5 Nov 2026 | ☑ Sent<br>☐ Not required<br>☐ Pending |
| **Sponsor device-deficiency route** | Does the signal point to a device, component, configuration or lot? | S-01 | GS-DD-014-0012, 28 Oct 2026 | ☑ Sent<br>☐ Not required<br>☐ Pending |
| **SART** | Are the individual events complete and routed through the safety pathway? | S-01 | SART-014-005; SART-014-006 opened, 30 Oct | ☑ Sent<br>☐ Not required<br>☐ Pending |
| **SRR** | Does the signal show a local process weakness to enter as a risk? | S-01 | SRR-002 reopened; SRR-006, 31 Oct | ☑ Sent<br>☐ Not required<br>☐ Pending |
| **CAPA log** | Did a first response fail, or does a process failure recur? | - | First occurrence; no failed response yet | ☐ Sent<br>☑ Not required<br>☐ Pending |
| **POT** | Does the PI need to restrict, supervise more, continue or escalate? | S-01 | POT review 3 Nov: S-03 proctored for 3 cases | ☑ Sent<br>☐ Not required<br>☐ Pending |
| **PIR / IMC** | Should the result inform an indicator or an internal check? | S-01 | IMP v1.1 CH-03: CSST monthly | ☑ Sent<br>☐ Not required<br>☐ Pending |

## Conclusion and sign-off

The conclusion reflects the most serious open signal.

| | |
|---|---|
| **Review conclusion** | ☐ No material pattern ☐ Closer surveillance ☑ Signal sent on ☐ Inconclusive: data limitation |
| **Next review · date and what will be checked again** | 30 Nov 2026, then monthly to January: instability by time point, liner +3 cases, S-03 cases, liner range in every set. |

**Limitations and rationale**

Very small numbers: 8 procedures, 2 participants at 3 months; the percentages move a lot with one event and no test of difference is attempted. The explanted liner has not yet been returned to the sponsor, so the device route has no result yet. ICDL-014-005 lot was read from the explant label.

Rationale: three triggers on instability point to the same few cases; device, technique and the new surgeon cannot be separated yet. Each route was sent so that each owner can test its own explanation.

**Reviewer**

Name / role: Maria Silva / Study Coordinator

Signature / date:

M. Silva 31 Oct 2026

**PI / accountable approver**

Name / role: Dr Alex Morgan / PI

Signature / date:

A. Morgan 3 Nov 2026